

第8讲. 微分学的应用.

例: P76. 1(2), (3), 2, 4, 5(1), 6, 7, 8.

P82. 1(1), (2), 2(1), 3.

一.

2. 曲面 $S: z = f(x, y) \in C^1(D)$

$$\triangleq F(x, y, z) = f(x, y) - z.$$

$$M_0(x_0, y_0, z_0), \quad P_0(x_0, y_0) \in D.$$

$$z_0 = f(p_0)$$

$$\vec{n} = \text{grad} F(M_0) = (f'_x(p_0), f'_y(p_0), -1)$$

$$S \text{ 在 } M_0 \text{ 切平面: } f'_x(p_0)(x-x_0) + f'_y(p_0)(y-y_0) - (z-z_0) = 0.$$

$$\text{--- 法线: } \frac{x-x_0}{f'_x(p_0)} = \frac{y-y_0}{f'_y(p_0)} = \frac{z-z_0}{-1}.$$

$$3. \text{ 曲面 } S \text{ 方程 } \begin{cases} x = x(u, v) \\ y = y(u, v) \\ z = z(u, v) \end{cases} \quad (u, v) \in D \subset \mathbb{R}^2.$$

$$\text{记号: } x, y, z \in C^1(D), \text{ 且 } \frac{\partial(x, y, z)}{\partial(u, v)} \text{ 秩为 } 2.$$

$$P_0(u_0, v_0) \in D$$

$$M_0(x_0, y_0, z_0) \in S.$$

$$\begin{cases} x_0 = x(p_0) \\ y_0 = y(p_0) \\ z_0 = z(p_0) \end{cases}$$

fix. v_0 .

$$\mathcal{L}_u: \begin{cases} x = x(u, v_0) \\ y = y(u, v_0) \\ z = z(u, v_0) \end{cases}$$

$$\mathcal{L}_u \text{ 在 } M_0 \text{ 切向量 } \vec{t}_u = (x'_u(p_0), y'_u(p_0), z'_u(p_0))$$

fix. u_0 .

$$\mathcal{L}_v: \begin{cases} x = x(u_0, v) \\ y = x(u_0, v) \\ z = z(u_0, v) \end{cases}$$

$$\mathcal{L}_v \text{ 在 } M_0 \text{ 切向量 } \vec{t}_v = (x'_v(p_0), y'_v(p_0), z'_v(p_0))$$

$$S \text{ 在 } M_0 \text{ 法向量 } \vec{n} = \vec{t}_u \times \vec{t}_v = \begin{vmatrix} i & j & k \\ x'_u & y'_u & z'_u \\ x'_v & y'_v & z'_v \end{vmatrix} \Big|_{P_0}$$

$$= \left(\det \frac{\partial(y, z)}{\partial(u, v)} \Big|_{P_0}, \det \frac{\partial(z, x)}{\partial(u, v)} \Big|_{P_0}, \det \frac{\partial(x, y)}{\partial(u, v)} \Big|_{P_0} \right)$$

$$\frac{\partial(x, y, z)}{\partial(u, v)} = \begin{pmatrix} x'_u & x'_v \\ y'_u & y'_v \\ z'_u & z'_v \end{pmatrix} \quad \text{秩为 } 2.$$

$$\text{曲面 } S \text{ 在 } M_0 \text{ 切平面: } A(x-x_0) + B(y-y_0) + C(z-z_0) = 0.$$

$$\text{--- 法线: } \frac{x-x_0}{A} = \frac{y-y_0}{B} = \frac{z-z_0}{C}$$

$$f: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3. \quad \text{连续可微}$$

$$P_0(u_0, v_0).$$

$$\Delta f = \begin{pmatrix} \Delta x \\ \Delta y \\ \Delta z \end{pmatrix}$$

$$\lim_{(\Delta u, \Delta v) \rightarrow 0} \frac{\|\Delta f - df(p_0)\|}{\|(\Delta u, \Delta v)\|} = 0.$$

$$\Delta f \approx df(p_0) = \frac{\partial(x, y, z)}{\partial(u, v)} \Big|_{p_0} \begin{pmatrix} \Delta u \\ \Delta v \end{pmatrix}$$

$$\begin{cases} x = x_0 + x'_u(p_0)(u-u_0) + x'_v(p_0)(v-v_0) \\ y = y_0 + y'_u(p_0)(u-u_0) + y'_v(p_0)(v-v_0) \\ z = z_0 + z'_u(p_0)(u-u_0) + z'_v(p_0)(v-v_0) \end{cases}$$

曲面 \$S\$ 在 \$M_0\$ 切平面参数方程.

$$z = \sqrt{x^2 + y^2} \quad \text{在 } (0,0) \text{ 不可微.}$$

例1. 设 \$f(x,y)\$ 可微 满足 \$\forall u, v, t. \quad f(tu, tv) = t^2 f(u, v)\$.

点 \$P(1, -1, 2)\$ 位于曲面 \$z = f(x, y)\$ 上. 设 \$f'_x(1, -1) = 3\$.

求曲面在点 \$P\$ 处切平面方程.

解: \$\vec{n} = (f'_x(1, -1), f'_y(1, -1), -1) = (3, \underline{f'_y(1, -1)}, -1)\$

$$z = f(1, -1).$$

$$tu f'_x(tu, tv) + tv f'_y(tu, tv) = 2t^2 f(u, v) = 2f(tu, tv)$$

$$x f'_x(x, y) + y f'_y(x, y) = 2f(x, y)$$

$$(x, y) = (1, -1).$$

$$3 - f'_y(1, -1) = 4. \quad \therefore f'_y(1, -1) = -1.$$

$$\therefore \vec{n} = (3, -1, -1)$$

$$\therefore \text{切平面: } 3(x-1) - (y+1) - (z-2) = 0.$$

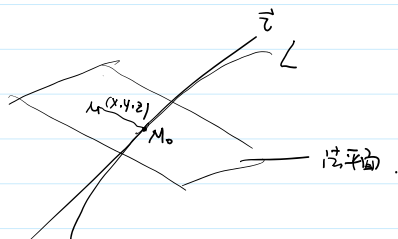
二. 曲线切线与法平面.

$$1. \text{ 光滑 } \angle: \begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases} \quad (t \in I) \quad t_0 \in I, \quad \begin{cases} x_0 = x(t_0) \\ y_0 = y(t_0) \\ z_0 = z(t_0) \end{cases}$$

$$\vec{T} = (x'(t), y'(t), z'(t)) \quad M_0(x_0, y_0, z_0) \in \angle.$$

$$\text{曲线 } \angle \text{ 在 } M_0 \text{ 切线: } \frac{x-x_0}{x'(t_0)} = \frac{y-y_0}{y'(t_0)} = \frac{z-z_0}{z'(t_0)}.$$

$$\therefore \text{法平面: } x'(t_0)(x-x_0) + y'(t_0)(y-y_0) + z'(t_0)(z-z_0) = 0.$$



2. $\mathcal{L}: \begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases}$ 曲线一般方程.

$F, G \in C^1 \quad M_0(x_0, y_0, z_0) \quad F(M_0) = 0, \quad G(M_0) = 0.$

$\frac{\partial(F, G)}{\partial(x, y, z)} \Big|_{M_0}$ 秩为 2.

||
 $\begin{pmatrix} F'_x & F'_y & F'_z \\ G'_x & G'_y & G'_z \end{pmatrix} \Big|_{M_0}$ $\det \frac{\partial(F, G)}{\partial(y, z)} \Big|_{M_0}$, $\det \frac{\partial(F, G)}{\partial(z, x)} \Big|_{M_0}$, $\det \frac{\partial(F, G)}{\partial(x, y)} \Big|_{M_0}$
 不同时为零.

不妨设 $\det \frac{\partial(F, G)}{\partial(y, z)} \Big|_{M_0} \neq 0.$

由 $\begin{cases} F(x, y, z) = 0 \\ G(x, y, z) = 0 \end{cases}$ 确定隐函数组 $\begin{cases} y = y(x) \\ z = z(x) \end{cases}$

曲线 $\mathcal{L}: \begin{cases} x = x \\ y = y(x) \\ z = z(x) \end{cases}$

$\vec{t} \perp \text{grad} F(M_0) \quad \vec{t} \perp \text{grad} G(M_0)$

$\mathcal{L}$ 在 M_0 切向量 $\vec{t} = \text{grad} F(M_0) \times \text{grad} G(M_0) = \begin{vmatrix} i & j & k \\ F'_x & F'_y & F'_z \\ G'_x & G'_y & G'_z \end{vmatrix} \Big|_{M_0}$

$= \left(\underbrace{\det \frac{\partial(F, G)}{\partial(y, z)} \Big|_{M_0}}_A, \underbrace{\det \frac{\partial(F, G)}{\partial(z, x)} \Big|_{M_0}}_B, \underbrace{\det \frac{\partial(F, G)}{\partial(x, y)} \Big|_{M_0}}_C \right)$

$\mathcal{L}$ 在 M_0 切线: $\frac{x-x_0}{A} = \frac{y-y_0}{B} = \frac{z-z_0}{C}$

... 法平面: $A(x-x_0) + B(y-y_0) + C(z-z_0) = 0.$

例2. 求 $\begin{cases} \frac{x^2}{4} + \frac{y^2}{4} + \frac{z^2}{2} = 1 \\ x - 2y + z = 0 \end{cases}$ 在 $P(1, 1, 1)$ 点切线及法平面.

解: $(\frac{x}{2}, \frac{y}{2}, z) \Big|_{(1, 1, 1)} = (\frac{1}{2}, \frac{1}{2}, 1)$

$\vec{n}_1 = (1, 1, 2)$

$\vec{n}_2 = (1, -2, 1)$

$\vec{t} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} i & j & k \\ 1 & 1 & 2 \\ 1 & -2 & 1 \end{vmatrix} = (5, 1, -3).$

$\therefore$ 切线: $\frac{x-1}{5} = \frac{y-1}{1} = \frac{z-1}{-3}$

法平面: $5(x-1) + y-1 - 3(z-1) = 0$.

三. 泰勒公式.

$f(x)$ 在 x_0 存在 n 阶导数.

$$f(x_0) + \underbrace{f'(x_0)(x-x_0)}_{\substack{\parallel \\ |x-x_0| \text{ 充分小.} \\ df(x_0)}} + \frac{1}{2} \underbrace{f''(x_0)(x-x_0)^2}_{\substack{\parallel \\ d^2f(x_0)}} + \dots + \frac{1}{n!} \underbrace{f^{(n)}(x_0)(x-x_0)^n}_{\substack{\parallel \\ d^nf(x_0)}}.$$

1. 高阶全微分.

$$z = f(x, y) \in C^2(D), \quad \forall (x_0, y_0) \in D, \quad \forall (x, y) \in D$$

$$\underbrace{df(x, y)}_{\text{波浪线}} = \underbrace{f'_x(x, y)dx + f'_y(x, y)dy}_{\text{波浪线}} \quad dx = h, \quad dy = k.$$

$$d^2f(x_0, y_0) = d(df(x, y))(x_0, y_0)$$

$$= (f''_{xx}(p_0)h + f''_{yx}(p_0)k)h + (f''_{xy}(p_0)h + f''_{yy}(p_0)k)k$$

$$= f''_{xx}(p_0)h^2 + 2f''_{xy}(p_0)hk + f''_{yy}(p_0)k^2 = \underbrace{(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y})^2}_{\text{波浪线}} f(p_0).$$

$$\underbrace{(a+b)^2}_{\text{波浪线}} = a^2 + 2ab + b^2$$

$$\underbrace{d^2f(x, y)}_{\text{波浪线}} = \underbrace{(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y})^2}_{\text{波浪线}} f(x, y)$$

$$d^3f(x_0, y_0) = d(d^2f(x, y))(x_0, y_0) = (h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y})^3 f(x_0, y_0)$$

$$\begin{array}{ccccc} & 1 & & 1 & \\ & & & & a+b \\ 1 & & 2 & & 1 \\ & 1 & & 3 & & 1 \\ 1 & & 3 & & 3 & & 1 \\ & 1 & & 4 & & 6 & & 4 & & 1 \end{array}$$

$$\text{设 } f(x, y) \in C^{n+1}(D).$$

$$\text{设 } d^n f(x, y) = (h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y})^n f(x, y)$$

$$= \sum_{i=0}^n C_n^i \frac{\partial^n f}{\partial x^i \partial y^{n-i}}(x, y) h^i k^{n-i}$$

$$\forall (x_0, y_0) \in D.$$

$$d(d^n f(x, y))(x_0, y_0) = d\left(\sum_{i=0}^n C_n^i \frac{\partial^n f}{\partial x^i \partial y^{n-i}}(x, y) h^i k^{n-i}\right)(x_0, y_0)$$

$$= \sum_{i=0}^n C_n^i d\left(\frac{\partial^n f}{\partial x^i \partial y^{n-i}}(x, y)\right)(x_0, y_0) h^i k^{n-i}$$

$$= \sum_{i=0}^n C_n^i \left(\frac{\partial^{n+1} f}{\partial x^{i+1} \partial y^{n-i}}(x_0, y_0) h + \frac{\partial^{n+1} f}{\partial x^i \partial y^{n-i+1}}(x_0, y_0) k \right) h^i k^{n-i}$$

$$= \sum_{i=0}^n C_n^{i+1} \frac{\partial^{n+1} f}{\partial x^{i+1} \partial y^{n-i}}(x_0, y_0) h^{i+1} k^{n-i} + C_n^n \frac{\partial^{n+1} f}{\partial x^n \partial y}(x_0, y_0) h^n.$$

$$\begin{aligned}
&= \sum_{i=1}^n C_n^{i-1} \frac{\partial^{n+1} f}{\partial x^i \partial y^{n+1-i}}(x_0, y_0) h^i k^{n+1-i} + C_n^n \frac{\partial^{n+1} f}{\partial x^{n+1}}(x_0, y_0) h^{n+1} \\
&+ \sum_{i=1}^n C_n^i \frac{\partial^{n+1} f}{\partial x^i \partial y^{n+1-i}}(x_0, y_0) h^i k^{n+1-i} + C_n^0 \frac{\partial^{n+1} f}{\partial y^{n+1}}(x_0, y_0) k^{n+1} \\
&= \sum_{i=0}^{n+1} C_{n+1}^i \frac{\partial^{n+1} f}{\partial x^i \partial y^{n+1-i}}(x_0, y_0) h^i k^{n+1-i} \\
&= \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^{n+1} f(x_0, y_0)
\end{aligned}$$

2. 泰勒公式:

定理1. 设 $D \subset \mathbb{R}^2$ 凸区域, $f(x, y) \in C^{n+1}(D)$. $P_0(x_0, y_0) \in D$.

则 $\forall P(x, y) \in D$, $f(x, y)$ 可在 P_0 展为带有拉格朗日余项的 $n+1$ 阶

泰勒公式:
$$f(x, y) = f(x_0, y_0) + (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}) f(x_0, y_0) + \frac{1}{2!} (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^2 f(x_0, y_0) + \dots + \frac{1}{n!} (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^n f(x_0, y_0) + \frac{1}{(n+1)!} (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^{n+1} f(x_0 + \theta h, y_0 + \theta k)$$

$$0 < \theta < 1.$$

$$h = x - x_0, \quad k = y - y_0.$$

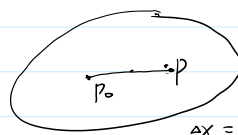
证: $\forall P(x, y) \in D$.

定义 $\overline{P_0 P} : \left\{ (x, y) \mid x = x_0 + t\Delta x, y = y_0 + t\Delta y, t \in [0, 1] \right\}$

$$t=0 \rightarrow P_0$$

$$t=1 \rightarrow P$$

$$\begin{cases} \Delta x = h \\ \Delta y = k \end{cases}$$



$$\Delta x = x - x_0$$

$$\Delta y = y - y_0$$

$$\overline{P_0 P} \subset D.$$

$$(\Delta x, \Delta y)$$

$$g(t) = f(x_0 + th, y_0 + tk) \in C^{n+1}[0, 1].$$

$$g(t) = g(0) + g'(0)t + \frac{1}{2!} g''(0)t^2 + \dots + \frac{1}{n!} g^{(n)}(0)t^n + \frac{1}{(n+1)!} g^{(n+1)}(\theta t) t^{n+1} \quad (0 < \theta < 1)$$

$$g(1) = f(x, y) \quad g(0) = f(x_0, y_0)$$

$$g(1) = g(0) + g'(0) + \frac{1}{2!} g''(0) + \dots + \frac{1}{n!} g^{(n)}(0) + \frac{1}{(n+1)!} g^{(n+1)}(\theta).$$

$$g'(0) = f'_x(x_0)h + f'_y(x_0)k = (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}) f(P_0).$$

$$g''(0) = \frac{d(g'(t))}{dt} \Big|_{t=0} = (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^2 f(P_0).$$

$$g'(t) = f'_x(x_0 + th, y_0 + tk) \cdot h + f'_y(x_0 + th, y_0 + tk) \cdot k$$

⋮

$$g^{(n)}(0) = (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^n f(P_0).$$

$$g^{(n+1)}(\theta) = (h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y})^{n+1} f(x_0 + \theta h, y_0 + \theta k) \quad 0 < \theta < 1.$$

$f(x, y)$ 在 $P_0(x_0, y_0)$ 处的 n 阶泰勒多项式 (唯一)

$$T_n = f(x_0, y_0) + f'_x(x_0, y_0)(x-x_0) + f'_y(x_0, y_0)(y-y_0) + \frac{1}{2} \left(f''_{xx}(x_0, y_0)(x-x_0)^2 + 2f''_{xy}(x_0, y_0)(x-x_0)(y-y_0) + f''_{yy}(x_0, y_0)(y-y_0)^2 \right) \\ + \dots + \frac{1}{n!} \sum_{i=0}^n C_n^i \frac{\partial^n f}{\partial x^i \partial y^{n-i}}(x_0, y_0) (x-x_0)^i (y-y_0)^{n-i}$$

含有 $\frac{\partial^n f}{\partial x^i \partial y^{n-i}}(x_0, y_0)$ 项 其系数为 $\frac{1}{n!} C_n^i (x-x_0)^i (y-y_0)^{n-i}$

例3. 求 $f(x, y) = (1+x)^{2y}$ 在 $(0, 1)$ 处的二阶泰勒多项式.

$$x^i (y-1)^{n-i}$$

令 $f(x, y) =$

$$(1+x)^{2y} = 1 + 2xy + \frac{1}{2} (2y)(2y-1)x^2 + o(x^2) \quad (x \rightarrow 0)$$

$$= 1 + 2x(y-1) + 2x + (y-1+1)[2(y-1)+1]x^2 + o(x^2)$$

$$= 1 + 2x(y-1) + 2x + x^2 + o(x^2 + |y-1|^2) \\ \left((x, y) \rightarrow (0, 1) \right)$$

$$\therefore T_2 = 1 + 2xy + x^2$$

例2.

$$f(x, y) = \begin{cases} \frac{1 - e^{x(x^2+y^2)}}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

求 $f(x, y)$ 在 $(0, 0)$ 处的四阶泰勒多项式, 并求 $f''_{xy}(0, 0)$, $\frac{\partial^4 f}{\partial x^4}(0, 0)$

解:

$$e^{x(x^2+y^2)} = 1 + x(x^2+y^2) + \frac{1}{2} x^2(x^2+y^2)^2 + o(x^2(x^2+y^2)^2) \quad (x, y) \rightarrow (0, 0)$$

$$\frac{1 - e^{x(x^2+y^2)}}{x^2+y^2} = -x - \frac{1}{2} x^2(x^2+y^2) + o(x^2(x^2+y^2))$$

$$T_4 = -x - \frac{1}{2} x^4 - \frac{1}{2} x^2 y^2$$

$$-\frac{1}{2} = \frac{1}{4!} \frac{\partial^4 f}{\partial x^4}(0, 0)$$

$$-\frac{1}{2} = \frac{1}{4!} \frac{\partial^4 f}{\partial x^2 \partial y^2}(0, 0) \cdot C_4^2$$